

IMO 2026 — Problem 1

Solution by Claude Fable 5

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Source: `results/claude-fable-5/problem-01/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\text{lcm}(m, n)}{\gcd(m, n)}$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius.

Solution document

Status

solved

Problem

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Approaches tried

- **Per-prime exponent dynamics + monovariant + gcd invariant** (see `approaches/potential-and-invariant`) — **successful, used in the final proof.** A move sends the exponent pair $(a, b) = (v_p(m), v_p(n))$ to $(\min(a, b), \max(a, b) - \min(a, b))$. Termination via the strictly decreasing potential $T + N$ ($T = \sum_i \Omega(x_i)$, $N = \text{number of entries} > 1$); part (b) via the invariance of $g_p = \gcd_i v_p(x_i)$ for each prime p .

- Computational verification (`code/check.py`): exhaustive BFS over the whole game tree for all multisets $\{a, b, c\} \subseteq [2, 14]$, 300 random 4-tuples in $[2, 30]$, 60 random 5-tuples in $[2, 12]$. Confirmed: every terminal state has exactly one entry > 1 , the surviving value is identical across all terminal states, and it equals $\prod_p p^{\gcd_i v_p(a_i)}$; also verified $T + N$ strictly decreases along every edge. All checks passed.

Current best

Complete proof of both parts. Explicit formula: if the initial numbers are $a_1, \dots, a_{2026}$, then the process always terminates and the unique surviving integer greater than 1 is

$$M = \prod_{p \text{ prime}} p^{\gcd(v_p(a_1), \dots, v_p(a_{2026}))},$$

which depends only on the initial numbers — proving (b). Part (a) follows from the strictly decreasing potential $T + N$ (termination) plus the fact that each move outputs at least one integer > 1 (so the count of entries > 1 never drops below 1).

Full proof

Throughout, the **state** of the blackboard is the tuple $(x_1, \dots, x_{2026})$ of the written numbers. For a prime p and a positive integer x , $v_p(x)$ denotes the exponent of p in x , and $\Omega(x) = \sum_p v_p(x)$ is the number of prime factors of x counted with multiplicity (so $\Omega(1) = 0$, and $\Omega(x) \geq 1$ for $x > 1$). We use the standard facts

$$v_p(\gcd(m, n)) = \min(v_p(m), v_p(n)), \quad v_p(\text{lcm}(m, n)) = \max(v_p(m), v_p(n)).$$

Lemma 1 (effect of a move on exponents)

Let a move replace $m, n > 1$ by $g = \gcd(m, n)$ and $h = \text{lcm}(m, n) / \gcd(m, n)$. Then g, h are positive integers, and for every prime p , writing $a = v_p(m)$, $b = v_p(n)$,

$$v_p(g) = \min(a, b), \quad v_p(h) = \max(a, b) - \min(a, b).$$

Proof. For every prime p we have $v_p(\gcd(m, n)) = \min(a, b) \leq \max(a, b) = v_p(\text{lcm}(m, n))$; hence $\gcd(m, n) \mid \text{lcm}(m, n)$, so h is a positive integer and $v_p(h) = \max(a, b) - \min(a, b)$. ■

All other entries are unchanged by the move, and every entry always remains a positive integer.

Lemma 2 (a move produces at least one entry > 1)

With the notation of Lemma 1, $\max(g, h) > 1$.

Proof. Suppose $h = 1$. Then $\text{lcm}(m, n) = \gcd(m, n)$. Since $\gcd(m, n) \mid m \mid \text{lcm}(m, n)$, this forces $g = \gcd(m, n) = m = \text{lcm}(m, n)$, and $m > 1$, so $g > 1$. Hence $h = 1$ implies $g > 1$; in all cases at least one of g, h exceeds 1. ■

Lemma 3 (monovariant)

For a state $S = (x_1, \dots, x_{2026})$ define

$$T(S) = \sum_{i=1}^{2026} \Omega(x_i), \quad N(S) = \#\{i : x_i > 1\}, \quad \Phi(S) = T(S) + N(S).$$

Then every move strictly decreases Φ , i.e. $\Phi(S') \leq \Phi(S) - 1$ for the state S' after the move.

Proof. Let the move replace m, n by g, h as above. Fix a prime p and let $a = v_p(m), b = v_p(n)$. By Lemma 1,

$$v_p(g) + v_p(h) = \min(a, b) + (\max(a, b) - \min(a, b)) = \max(a, b) = a + b - \min(a, b) = v_p(m) + v_p(n) - v_p(g).$$

Summing over all primes p gives $\Omega(g) + \Omega(h) = \Omega(m) + \Omega(n) - \Omega(g)$, hence

$$T(S') = T(S) - \Omega(g).$$

For N : the two chosen positions contribute 2 to $N(S)$ (both $m, n > 1$) and at most 2 to $N(S')$; all other positions are unchanged. Hence $N(S') \leq N(S)$ always. We distinguish two cases.

Case $g = 1$. Then $h = \text{lcm}(m, n) = mn / \gcd(m, n) = mn > 1$, so the two positions contribute exactly 1 to $N(S')$, giving $N(S') = N(S) - 1$; and $T(S') = T(S) - \Omega(1) = T(S)$. Thus $\Phi(S') = \Phi(S) - 1$.

Case $g > 1$. Then $\Omega(g) \geq 1$, so $T(S') \leq T(S) - 1$, and $N(S') \leq N(S)$. Thus $\Phi(S') \leq \Phi(S) - 1$.

In both cases Φ decreases by at least 1. ■

Part (a)

Termination. Φ is a nonnegative integer for every state (each $x_i \geq 1$), and by Lemma 3 it strictly decreases with every move. Hence at most $\Phi(S_0)$ moves can be made from the initial state S_0 : after finitely many moves the process stops.

The process stops exactly when $N \leq 1$. A move requires two entries > 1 in different places, and any two entries > 1 can be chosen; thus a move is possible if and only if $N \geq 2$. So the final state satisfies $N \leq 1$.

$N \geq 1$ at all times. Initially $N = 2026 \geq 1$. Consider any move, performed from a state with $N \geq 2$. The two chosen positions held two entries > 1 and, by Lemma 2, hold at least one entry > 1 afterwards; all other positions are unchanged. Hence N decreases by at most 1 per move, so after the move $N \geq 2 - 1 = 1$. By induction, $N \geq 1$ in every state ever reached.

Combining: the process terminates after finitely many moves, and in the final state $1 \leq N \leq 1$, i.e. **exactly one integer on the blackboard is greater than 1**. Call it M . This proves (a). ■

Part (b)

For a state $S = (x_1, \dots, x_{2026})$ and a prime p define

$$g_p(S) = \gcd(v_p(x_1), \dots, v_p(x_{2026})) \in \mathbb{Z}_{\geq 0},$$

with the standard conventions $\gcd(t, 0) = t$ and $\gcd(0, 0, \dots, 0) = 0$ (so $g_p(S) = 0$ exactly when p divides no entry).

Lemma 4 (invariance of g_p)

For every prime p and every move $S \rightarrow S'$ we have $g_p(S') = g_p(S)$.

Proof. First, for all integers $a, b \geq 0$,

$$\gcd(\min(a, b), \max(a, b) - \min(a, b)) = \gcd(a, b). \quad (*)$$

Indeed, WLOG $a \leq b$; an integer d divides both a and $b - a$ if and only if it divides both a and b , so the pairs $(a, b - a)$ and (a, b) have the same common divisors and hence the same gcd. (If $a = b = 0$, both sides are 0.)

Now let the move act on entries m, n with $a = v_p(m)$, $b = v_p(n)$, and let r denote the gcd of the values $v_p(x_i)$ over all the *other* 2024 positions. By Lemma 1 and the associativity of gcd,

$$g_p(S') = \gcd\left(r, \gcd(\min(a, b), \max(a, b) - \min(a, b))\right) \stackrel{(*)}{=} \gcd(r, \gcd(a, b)) = g_p(S). \quad \blacksquare$$

Conclusion of (b). Let $S_0 = (a_1, \dots, a_{2026})$ be the initial state and let S_f be any final state reached by any sequence of choices. By part (a), S_f consists of one integer $M > 1$ and 2025 ones. Hence for every prime p ,

$$g_p(S_f) = \gcd(v_p(M), 0, \dots, 0) = v_p(M).$$

By Lemma 4, $g_p(S_f) = g_p(S_0)$ for every prime p . Therefore

$$v_p(M) = g_p(S_0) = \gcd(v_p(a_1), \dots, v_p(a_{2026})) \quad \text{for every prime } p,$$

and since only the finitely many primes dividing $a_1 \cdots a_{2026}$ give a nonzero exponent,

$$M = \prod_{p \text{ prime}} p^{\gcd(v_p(a_1), \dots, v_p(a_{2026}))}.$$

This expression depends only on the initial numbers $a_1, \dots, a_{2026}$ and not on any of Confucius's choices. Hence the value of M is the same for every legal sequence of moves. $\blacksquare$

(Consistency check: since $a_1 > 1$, some prime p satisfies $v_p(a_1) \geq 1$, whence $g_p(S_0) \geq 1$ and indeed $M > 1$, matching part (a).)